

# Junyan Zhang

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## CONTACT INFORMATION

Email: [yx3x@ustc.edu.cn](mailto:yx3x@ustc.edu.cn), [zhangjy9610@gmail.com](mailto:zhangjy9610@gmail.com)  
Affiliation: School of Mathematical Sciences, University of Science and Technology of China. Address: 96 Jinzhai Road, Baohe District, Hefei, Anhui 230026, China.  
Personal webpage: <https://www.zhangjy9610.me>

## EMPLOYMENT

### University of Science and Technology of China (USTC)

Specially-Appointed Professor (Tenure-Track). May 26, 2025 ~ Now.

### National University of Singapore (NUS)

Peng Tsu Ann Assistant Professor (Postdoc). August 1, 2022 ~ May 14, 2025.

- Mentor: Professor Yao Yao.

## EDUCATION

### Johns Hopkins University (JHU)

Ph.D. in Mathematics. August 27, 2017 ~ May 22, 2022.

- Dissertation: The Free-Boundary Problems in Inviscid Magnetohydrodynamics with or without Surface Tension.
- Advisor: Professor Hans Lindblad.

### University of Science and Technology of China (USTC)

B.Sc. in Mathematics. August 14, 2013 ~ June 21, 2017.

- Thesis: Inviscid damping and Asymptotic Stability of PDEs in fluids.
- Advisor: Professor Lifeng Zhao.

## RESEARCH INTERESTS

I study fluid PDEs. Currently I focus on:

- Nonlinear stability of compressible vortex sheets and contact discontinuities in various kinds of fluids with or without surface tension.
- Singular limits of various types of fluid PDEs, multi-scale singular limits.
- Singularity or stability or long-time evolution of fluid PDEs.

## PUBLICATIONS & PREPRINTS

16. Zhile Li, Junyan Zhang, Lifeng Zhao. *Linear Stability and Inviscid Damping of Monotone Shear Flows for 2D Compressible Euler Equations*. [arxiv:2608.27018](#), preprint.
15. Qiangchang Ju, Jiawei Wang, Junyan Zhang. *Uniform Anisotropic Regularity and Low Mach Number Limit of Non-isentropic Ideal MHD Equations with a Perfectly Conducting Boundary*. [arxiv:2412.09943](#), preprint.
14. Jiawei Wang, Junyan Zhang. *Low Mach Number Limit of Non-isentropic Inviscid Elastodynamics with General Initial Data*. [arxiv:2412.09941](#), preprint.
13. Junyan Zhang. *On the Incompressible Limit of Current-Vortex Sheets with or without Surface Tension*. [arxiv:2405.00421](#), preprint.
12. Junyan Zhang. *Well-posedness and Incompressible Limit of Current-Vortex Sheets with Surface Tension in Ideal Compressible MHD*. [arxiv:2312.11254v3](#), preprint.
11. Jiawei Wang, Junyan Zhang. *Incompressible Limit of Compressible Ideal MHD Flows inside a Perfectly Conducting Wall*. **J. Differ. Equ.**, Vol.425, 846-894 (2025).
10. Chenyun Luo, Junyan Zhang. *Compressible Gravity-Capillary Water Waves with Vorticity: Local Well-posedness, Incompressible and Zero-Surface-Tension Limits*. **J. Differ. Equ.**, Vol. 472, 114444 (2026), 114 pages.

9. Xumin Gu, Chenyun Luo, Junyan Zhang. *Zero Surface Tension Limit of the Free-Boundary Problem in Incompressible Magnetodynamics*. **Nonlinearity**, 35(12), 6349-6398 (2022).
8. Hans Lindblad, Junyan Zhang. *Anisotropic Regularity of the Free-Boundary Problem in Compressible Ideal Magnetohydrodynamics*. **Arch. Rational Mech. Anal.**, 247(5), no. 89: 1-94 (2023).
7. Xumin Gu, Chenyun Luo, Junyan Zhang. *Local Well-posedness of the Free-Boundary Incompressible Magnetohydrodynamics with Surface Tension*. **J. Math. Pures Appl.**, Vol. 182, 31-115, (2024).
6. Junyan Zhang. *Local Well-posedness and Incompressible Limit of the Free-Boundary Problem in Compressible Elastodynamics*. **Arch. Rational Mech. Anal.**, 244(3), 599-697 (2022).
5. Junyan Zhang. *Local Well-posedness of the Free-Boundary Problem in Compressible Resistive Magnetohydrodynamics*. **Calc. Var. Partial Differ. Equ.**, 62(4), no.124: 1-60 (2023).
4. Chenyun Luo, Junyan Zhang. *Local Well-posedness for the Motion of a Compressible Gravity Water Wave with Vorticity*. **J. Differ. Equ.**, Vol. 332, 333-403 (2022).
3. Junyan Zhang. *A priori Estimates for the Free-Boundary Problem of Compressible Resistive MHD Equations and Incompressible Limit*. **arxiv: 1911.04928** preprint.
2. Chenyun Luo, Junyan Zhang. *A priori Estimates for the Incompressible Free-Boundary Magnetohydrodynamics Equations with Surface Tension*. **SIAM J. Math. Anal.**, 53(2), 2595-2630 (2021).
1. Chenyun Luo, Junyan Zhang. *A Regularity Result for the Incompressible Magnetohydrodynamics Equations with Free Surface Boundary*. **Nonlinearity**, 33(4), 1499-1527 (2020).

REFeree SERVICES So far I have been a reviewer for the following journals.

- Cambridge J. Math., Arch. Rational Mech. Anal., Commun. Math. Phys., SIAM J. Math. Anal., Nonlinearity, J. Differ. Equ., J. Math. Phys., DCDS.

INVITED TALKS AND  
MINI-COURSES

- *Incompressible limit of free-surface inviscid fluids*:  
The 13-th PDE Youth Forum, Dalian University of Technology, May 09 2026;  
Workshop on Nonlinear PDEs in Fluid Mechanics, AMSS, Chinese Academy of Sciences, Aug 13 2026;  
Institute of Applied Physics and Computational Mathematics (Beijing), Aug 14 2026;  
The 34-th Yangzi River PDE Conference, Fudan University, Aug 23 2026.
- *Incompressible limit of non-isentropic ideal MHD with a perfectly conducting boundary*:  
Institute of Applied Mathematics, AMSS, Chinese Academy of Sciences, July 15 2024, Dec 02 2024;  
1st Asian-Pacific International Conference on Dispersive Equations (APICDE), Jimei University, Oct 30 2024;  
School of Mathematical Sciences, Peking University, Dec 02 2024;  
Workshop on “Old and New Challenges in Fluid Equations”, National University of Singapore, Dec 19 2024;  
Institute of Mathematics and Interdisciplinary Sciences, Nanchang University, Jan 08 2025.
- *Well-posedness and incompressible limit of current-vortex sheets in ideal compressible MHD*:  
National University of Singapore, Nov 1 2023;  
University of Science and Technology of China, Dec 5 2023;  
Institute of Applied Physics and Computational Mathematics (Beijing), Dec 19 2023;  
Morningside Center of Mathematics, Chinese Academy of Sciences, Dec 20 2023;  
Institute of Mathematical Sciences, The Chinese University of Hong Kong, May 22 2024;  
The 19-th international conference of “Hyperbolic problems: theory, numerics and appli-

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|                                  | <p>cations”, Shanghai Jiao Tong University, July 2 2024;<br/>Hangzhou Normal University, Oct 14 2025.</p> <ul style="list-style-type: none"> <li>• <i>The Free-boundary Problems in Inviscid Compressible Fluids (Mini-Course)</i>:<br/>School of Mathematical Sciences, University of Science and Technology of China, June 28–29 2023;<br/>Institute of Mathematics and Interdisciplinary Sciences, Nanchang University, June 4–6 2024.</li> <li>• <i>On the motion of compressible gravity-capillary water waves with vorticity</i>:<br/>PDE &amp; Scientific computing seminar, National University of Singapore, Aug 26 2022;<br/>Institute of Mathematical Sciences, The Chinese University of Hong Kong, May 19 2023;<br/>Institute of Applied Physics and Computational Mathematics (Beijing), Apr 12 2024.</li> <li>• <i>Local well-posedness of incompressible ideal MHD with surface tension</i>. PDE seminar organised by Looi Shi-Zhuo, March 23 2022.</li> <li>• <i>Anisotropic regularity of free-boundary compressible ideal MHD</i>:<br/>Institute of Mathematical Sciences, The Chinese University of Hong Kong, Oct 14 2021;<br/>Analysis &amp; PDE Seminar, UC Berkeley, Oct 25 2021;<br/>PDE Seminar, Vanderbilt University, Nov 5 2021;<br/>Analysis of Fluids Seminar, Princeton University, Feb 17 2021.</li> <li>• <i>Local well-posedness and incompressible limit of free-boundary compressible elastodynamics</i>, Webinar on APDE, June 5 2021.</li> <li>• <i>Local well-posedness and incompressible limit of the free-boundary compressible resistive MHD equations</i>, Wuhan University, Jan 10 2021.</li> <li>• <i>Local well-posedness for the motion of compressible gravity water wave</i>, University of Science and Technology of China, Nov 6 2020.</li> <li>• <i>On the free-boundary problem of MHD equations with or without surface tension</i>, University of Science and Technology of China, Dec 23 2019.</li> <li>• <i>On the Incompressible MHD with or without Surface Tension</i>, Institute of Mathematics, Chinese Academy of Sciences, May 23 2019.</li> </ul> |
| ACADEMIC VISITING                | <ul style="list-style-type: none"> <li>• The Chinese University of Hong Kong. May 10 2023-June 10 2023, May 13 2024-May 25 2024.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CONFERENCES & WORKSHOPS ATTENDED | <ul style="list-style-type: none"> <li>• <i>The 34-th Yangzi River PDE Conference</i>, Fudan University, Aug 22-24 2026.</li> <li>• <i>Workshop on Nonlinear PDEs in Fluid Mechanics</i>, AMSS, Chinese Academy of Sciences, Aug 12-14 2026;</li> <li>• <i>The 13-th PDE Youth Forum, Dalian University of Technology</i>, May 08-10 2026;</li> <li>• <i>Workshop on “Old and New Challenges in Fluid Equations”</i>, Singularities in fluids and general relativity, National University of Singapore, Dec 16-20 2024.</li> <li>• <i>The 1st Asian-Pacific International Conference on Dispersive Equations (APICDE)</i>, Jimei University, Oct 27-Nov 1 2024.</li> <li>• <i>The 19-th international conference of “Hyperbolic problems: theory, numerics and applications”</i>, Shanghai Jiao Tong University, July 1-July 5 2024.</li> <li>• <i>Mathematics of Fluid Dynamics program</i>, UC Berkeley MSRI, Jan-May 2021 (online due to the COVID-19 pandemic).</li> <li>• <i>Long Time Behavior and Singularity Formation in PDEs</i>, New York University Abu Dhabi, May 2020 and Dec 2020 (online due to the COVID-19 pandemic).</li> <li>• <i>2019 Southern California Analysis and PDE Conference</i>, UCSD, November 2019.</li> <li>• <i>Summer School on Mathematical General Relativity and the Geometric Analysis of Waves of Fluids</i>, MIT, June 2018.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| TEACHING                         | <p><b>University of Science and Technology of China (Instructor)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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|-------------|-----------------------------------------------------------------|
| 2027 Spring | Nonlinear Hyperbolic PDEs (Gradute)                             |
| 2026 Fall   | Ordinary Differential Equations, Partial Differential Equations |
| 2026 Summer | Frontier Lectures on Mathematics                                |
| 2026 Spring | Modern Partial Differential Equations (Graduate)                |
| 2025 Fall   | Partial Differential Equations, Freshmen Seminar                |

#### **National University of Singapore (Instructor)**

|             |                                                    |
|-------------|----------------------------------------------------|
| 2025 Spring | Advanced Partial Differential Equations (Graduate) |
| 2024 Spring | Partial Differential Equations                     |
| 2023 Spring | Partial Differential Equations                     |

#### **Johns Hopkins University (TA/Grader)**

|             |                                                      |
|-------------|------------------------------------------------------|
| 2022 Spring | Honor Analysis II, ODE                               |
| 2021 Fall   | Introduction to Proofs, Graduate Real Analysis       |
| 2020 Fall   | Honor Analysis I, ODE                                |
| 2020 Spring | Honor Analysis II, Undergrad PDE                     |
| 2019 Fall   | Honor Analysis I, Graduate Real Analysis             |
| 2019 Spring | Honor Analysis II, Calculus II (Engineering)         |
| 2018 Fall   | Calculus II (Engineering)                            |
| 2018 Spring | Undergrad PDE                                        |
| 2017 Fall   | Undergrad Complex Analysis, Calculus I (Engineering) |

#### **University of Science and Technology of China (TA)**

|             |                                      |
|-------------|--------------------------------------|
| 2017 Spring | Differential Equations II (Graduate) |
| 2016 Fall   | Advanced Real Analysis (Graduate)    |
| 2016 Spring | Honor Real Analysis                  |